

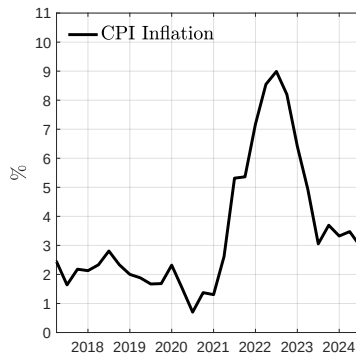
# The Inflation Attention Threshold and Inflation Surges

Oliver Pfäuti  
UT Austin

32nd Annual Symposium  
Society for Nonlinear Dynamics and Econometrics  
March 2025

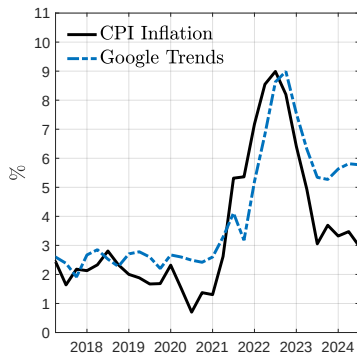
# Inflation is back...

(a) Inflation



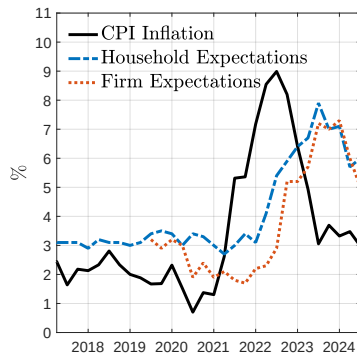
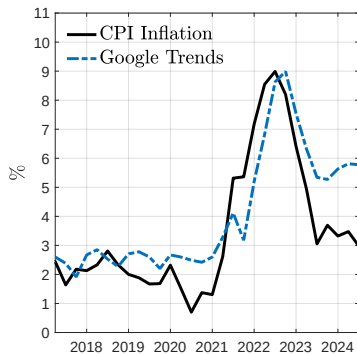
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(a) Inflation and Google Trends



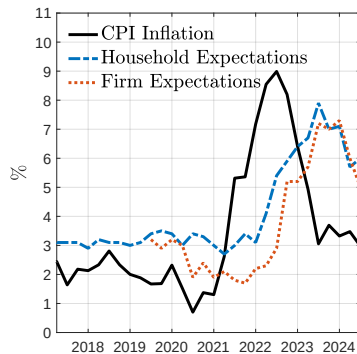
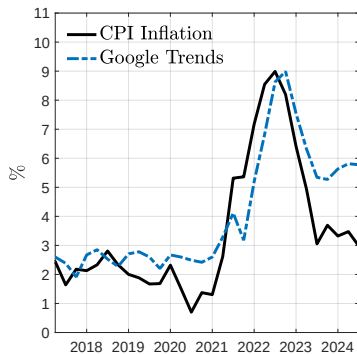
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(a) Inflation and Google Trends      (b) Inflation and expectations



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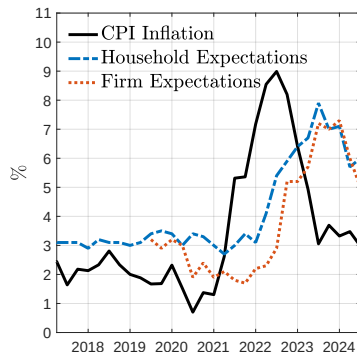
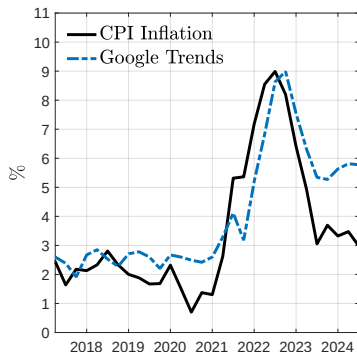
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**Q:** When does attention to inflation change?

**Q:** Can attention changes explain inflation & inflation expectations dynamics?

# Preview

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  - ▶ exceeding threshold changes inflation dynamics (resembling recent inflation surge)
  - ▶ threshold leads to inflation asymmetry, longer 'last mile', larger central bank losses, ...

# Contribution to the literature

- ▶ **Drivers of recent inflation surge:** Shapiro (2023), Gagliardone/Gertler (2023), Bernanke/Blanchard (2023), Benigno/Eggertsson (2023), Amiti et al. (2023), Bianchi/Melosi (2022) & Bianchi et al. (2023), Reis (2022), Erceg et al. (2024)...  
⇒ Contribution: role of attention increase in inflation surge
- ▶ **Measuring attention to inflation:** Cavallo et al. (2017), Pfäuti (2021), Link et al. (2024), Korenok et al. (2022), Bracha/Tang (2023), Weber et al. (2023), Kroner (2023)  
⇒ Contribution: estimate attention threshold and attention in a way that directly maps into otherwise standard macro models
- ▶ **State dependency of shocks:** Auerbach/Gorodnichenko (2012a,b), Ramey/Zubairy (2018), Jo/Zubairy (2023), Tenreyro/Thwaites (2016), Ascari/Haber (2022), Joussier et al. (2023)  
⇒ Contribution: role of attention regime for inflation response
- ▶ **Theory:** Mackowiak/Wiederholt (2009), Paciello/Wiederholt (2014), Reis (2006a,b), Pfäuti (2021), Carvalho et al. (2022), Afrouzi/Yang (2022), Gati (2022)  
⇒ Contribution: GE model with attention threshold, role for inflation surges

# Outline

1. Quantify Attention and Attention Threshold
2. Role of Attention for Inflation
3. Model + Model Results



## Quantifying attention to inflation

- ▶ Perceived law of motion (subscripts  $r$  indicate potential regime dependence):

$$\pi_t = (1 - \rho_{\pi,r})\underline{\pi}_r + \rho_{\pi,r}\pi_{t-1} + \nu_t, \text{ with } \nu_t \sim N(0, \sigma_\nu^2)$$

- ▶ current inflation is unobservable
- ▶ noisy signal:  $s_t = \pi_t + \varepsilon_t$ , with  $\varepsilon_t \sim N(0, \sigma_{\varepsilon,r}^2)$ , precision  $\frac{1}{\sigma_{\varepsilon,r}^2}$  reflects attention

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$$\tilde{E}_t\pi_{t+1} = (1 - \rho_{\pi,r})\underline{\pi}_r + \rho_{\pi,r}\tilde{E}_{t-1}\pi_t + \rho_{\pi,r}\gamma_{\pi,r} \left( \pi_t - \tilde{E}_{t-1}\pi_t \right) + u_t$$

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$$\tilde{E}_t\pi_{t+1} = \beta_{0,r} + \beta_{1,r}\tilde{E}_{t-1}\pi_t + \beta_{2,r} \left( \pi_t - \tilde{E}_{t-1}\pi_t \right) + \epsilon_t,$$

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Rational inattention microfoundation: ▶ Details

$$\gamma_{\pi,r} = \max \left( 0, 1 - \frac{\lambda_r}{\rho_{\pi,r}^2 \sigma_{\pi,r}^2} \right)$$

# Attention threshold

- ▶ Test for different **attention levels** and **attention threshold**  $\bar{\pi}$ :

$$\begin{aligned}\tilde{E}_t \pi_{t+1} = & \mathbb{1}_{\pi_{t-1} \leq \bar{\pi}} \left( \beta_{0,L} + \beta_{1,L} \tilde{E}_{t-1} \pi_t + \beta_{2,L} \left( \pi_t - \tilde{E}_{t-1} \pi_t \right) \right) \\ & + (1 - \mathbb{1}_{\pi_{t-1} \leq \bar{\pi}}) \left( \beta_{0,H} + \beta_{1,H} \tilde{E}_{t-1} \pi_t + \beta_{2,H} \left( \pi_t - \tilde{E}_{t-1} \pi_t \right) \right) + \tilde{\epsilon}_t\end{aligned}$$

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- ▶ Estimate threshold  $\bar{\pi}$  and regression coefficients jointly by minimizing SSR
- ▶ Baseline data:
  - ▶ monthly average expectations Michigan Survey of Consumers, 1978-2024
  - ▶ actual inflation: U.S. CPI inflation ▶ Time series



Empirical results: attention twice as high when inflation is above 4%

|          | Threshold $\bar{\pi}$ | Low Att. $\hat{\gamma}_{\pi,L}$ | High Att. $\hat{\gamma}_{\pi,H}$ | $p$ -val. $\gamma_{\pi,L} = \gamma_{\pi,H}$ |
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| Baseline | 3.91%                 | 0.18                            | 0.35                             | 0.000                                       |
| s.e.     |                       | (0.018)                         | (0.042)                          |                                             |

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- ▶ robustness:
  - ▶ no evidence for changes within regime and data favors having one threshold
  - ▶ regional data, control for unemployment expectations
  - ▶ median expectations, quarterly frequency, NY Fed SCE (HH panel), SPF
  - ▶ threshold defined by inflation level yields better fit than volatility (salience?)
  - ▶ Markov-switching model

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2. **Role of Attention for Inflation**
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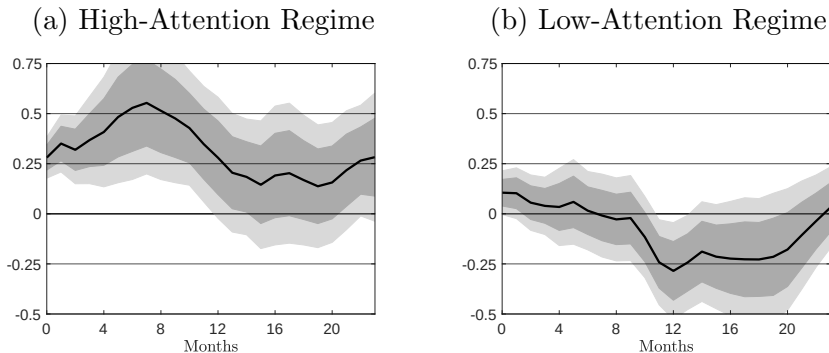
# Attention regimes and the propagation of supply shocks

Estimate local projection:

$$y_{i,t+j} = \mathbb{1}_{i,H} (\alpha_{i,j}^H + \beta_j^H \varepsilon_t) + (1 - \mathbb{1}_{i,H}) (\alpha_{i,j}^L + \beta_j^L \varepsilon_t) + \Gamma' X_{i,t} + u_{i,t+j}$$

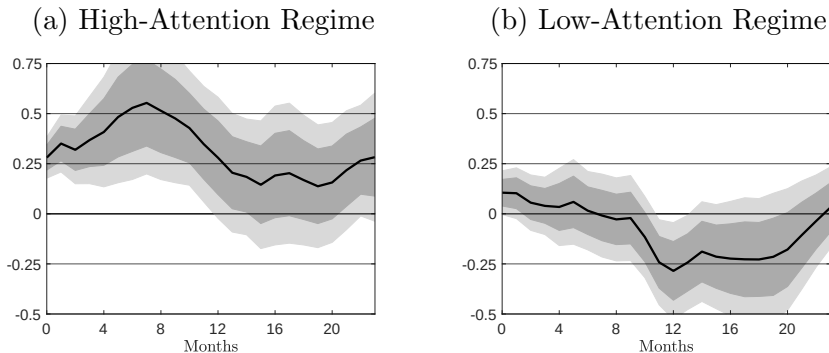
- ▶  $y_{i,t+j}$  : y-o-y CPI inflation in period  $t + j$  in region  $i$
- ▶  $\mathbb{1}_{i,H} = 1$  if region  $i$  is in high-attention regime (based on Google Trends)
- ▶  $\varepsilon_t$  : oil supply news shock, 1975M1-2023M12 (Känzig, AER 2021)
- ▶  $\beta_j^r$  : effect of supply shock on inflation at horizon  $j$  in regime  $r \in \{L, H\}$
- ▶  $X_{i,t}$ : controls, including (lagged) aggregate inflation, regional inflation & unemployment, and interaction of those with the shock  
(robustness: control for expectations about other variables)

# Supply shocks are more inflationary in times of higher attention



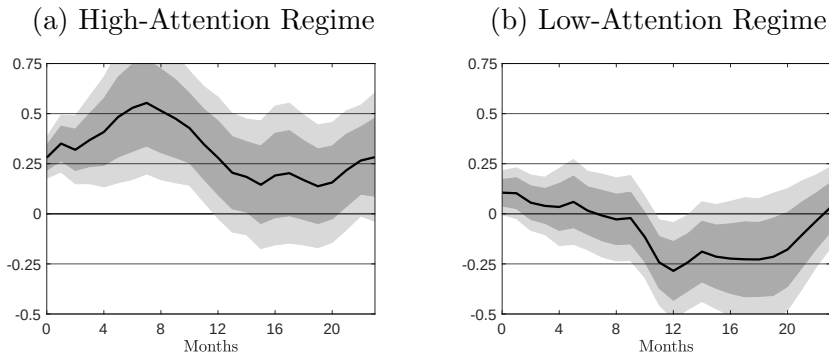
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- ▶ also note different shape: hump vs. peak on impact
- ▶ robust to interacting Google trends with the shock & controlling for time FEs

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# Model sketch

New Keynesian model [with limited attention and attention threshold](#): [► Details](#)

- [Households](#): consume, work, [subjective expectations + limited attention](#) [► Details](#)

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  - ▶ **Intermediate producers:** monopolistic competition, price adjustment cost, use labor to produce, tax/subsidy, **subjective expectations + limited attention** ▶ Details

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- ▶ [Government](#):
  - ▶ [Fiscal authority](#): subsidy to firms, lump-sum taxes, issues bonds [► Details](#)
  - ▶ [Monetary authority](#): sets nominal interest rate, following Taylor rule

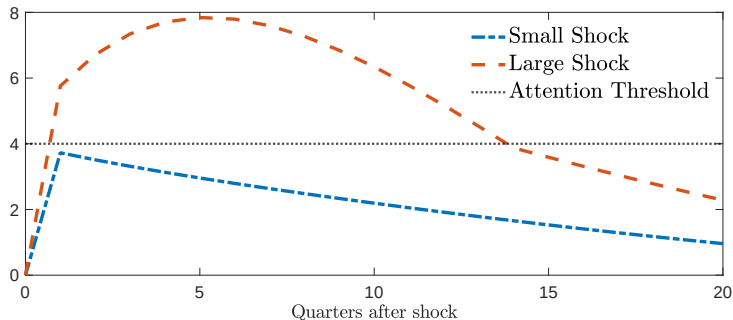
# Cost-push shock: attention shift alters inflation dynamics

- ▶ Effects of cost-push shocks  $u_t$  on inflation?
  - ▶ Equilibrium
  - ▶ Analytical Example
- 1. large shock that pushes inflation above the threshold
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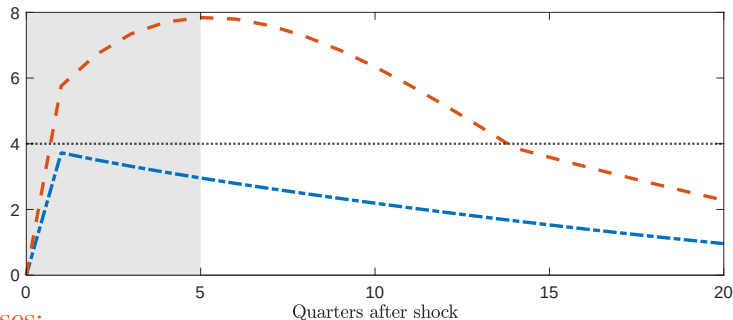
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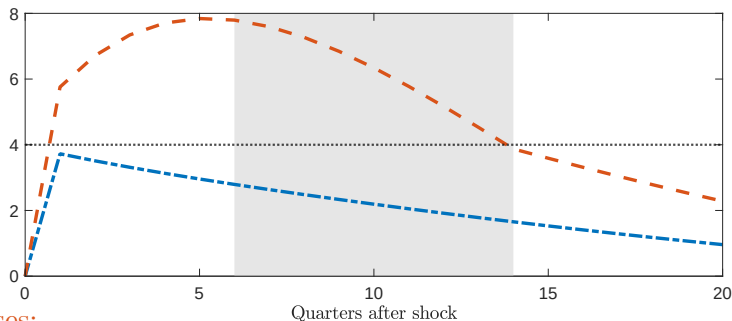
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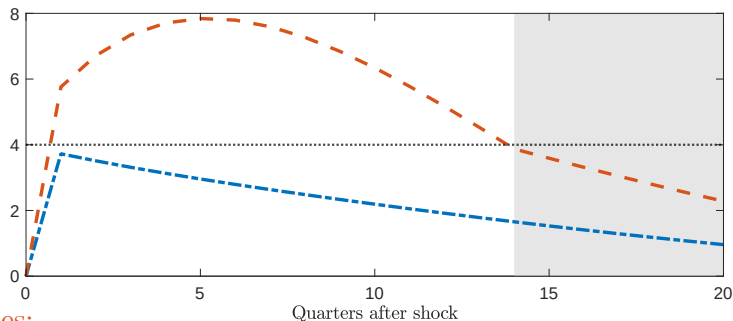


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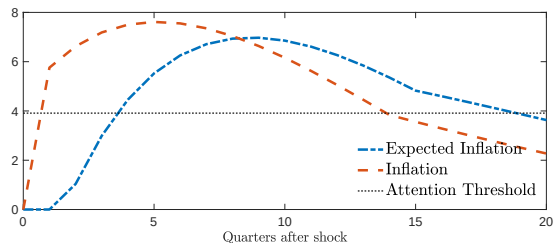
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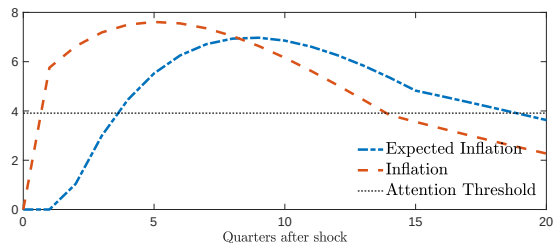
- **Three phases:**
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  2. relatively fast disinflation initially due to shock dying out and high attention
  3. **disinflation slows down** once inflation falls back **below threshold**

# Inflation and inflation expectation dynamics: Model vs. Data

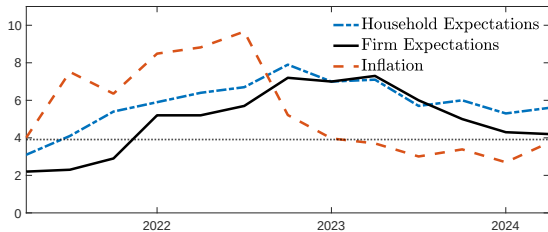


- **Model:** inflation hump-shaped and inflation expectations initially undershoot, followed by delayed overshooting

# Inflation and inflation expectation dynamics: Model vs. Data



- **Model:** inflation hump-shaped and inflation expectations initially undershoot, followed by delayed overshooting



- **Data:** shows similar patterns

# Additional Results

- ▶ Similar results for demand shocks ▶ IRFs
- ▶ Attention threshold induces asymmetry in inflation dynamics: thicker right tail
  - ▶ Asymmetry
- ▶ Dovish monetary policy rules lead to larger central bank losses due to... ▶ Details
  - ... higher inflation volatility
  - ... positive average inflation due to asymmetry

# Conclusion

- ▶ Recent inflation surge brought inflation back on people's minds
- ▶ I find that...
  - ... attention doubles once inflation exceeds 4%
  - ... attention amplifies supply shocks and played important role in recent inflation surge
  - ... changes in attention matter for dynamics of inflation and inflation expectations
  - ... dovish monetary policy may lead to substantial central bank losses

# Appendix

# Limited-Attention Model

Model of **optimal attention choice**:

- ▶ Agent (household or firm) needs to form an expectation about future inflation
- ▶ Acquiring information is costly (cognitive abilities, time, etc.)
- ▶ Making mistakes leads to utility losses

⇒ optimal level of attention depends on how costly information acquisition is, how high your stakes are and the properties of inflation itself



# Setup

Agent believes that inflation follows an AR(1) process:

$$\pi' = \rho_{\pi}\pi + \nu,$$

with  $\rho_{\pi} \in [0, 1]$  and  $\nu \sim i.i.N(0, \sigma_{\nu}^2)$ .

The full-information forecast is given by

$$\pi^{e*} = \rho_{\pi}\pi$$

**Problem:** current inflation is unobservable and acquiring information is costly.

# Information Acquisition Problem

The agent's problem:

- ▶ Choose the form of the signal  $s$
- ▶ to minimize the loss that arises from making mistakes,  $U(s, \pi)$
- ▶ facing the cost of information  $C(f) = \lambda I(\pi; s)$ , with  $I(\pi; s)$  being the expected reduction in entropy of  $\pi$  due to observing  $s$

# Information Acquisition Problem Continued

Quadratic loss function

$$U(\pi^e, \pi) = r \left( \underbrace{\rho_\pi \pi}_{\text{full-info}} - \pi^e \right)^2$$

$r$ : stakes

Optimal signal has the form (Matejka/McKay (2015))

$$s = \pi + \varepsilon$$

where  $\varepsilon \sim i.i.N(0, \sigma_\varepsilon^2)$  captures noise

$\sigma_\varepsilon^2$  is chosen optimally

## Optimal Level of Attention

The optimal forecast is given by

$$\pi^e = \rho_\pi \hat{\pi} + \rho_\pi \gamma (s - \hat{\pi}),$$

where  $\hat{\pi}$  is the prior belief of the agent and  $\gamma$  is the **optimal level of attention**:

$$\gamma = \max \left( 0, 1 - \frac{\lambda}{2r\rho_\pi^2\sigma_\pi^2} \right)$$

Attention is higher when:

- ▶ the cost of information  $\lambda$  is low
- ▶ the stakes  $r$  are high
- ▶ inflation is very volatile (high  $\sigma_\pi^2$ ) or persistent (high  $\rho_\pi$ ) ▶ [Back](#)

# Households

Representative household, lifetime utility:

$$\tilde{E}_0 \sum_{t=0}^{\infty} \beta^t Z_t \left[ \frac{C_t^{1-\sigma}}{1-\sigma} - \Xi H_t \right]$$

Households maximize their lifetime utility subject to the flow budget constraints

$$C_t + B_t = w_t H_t + \frac{1 + i_{t-1}}{1 + \pi_t} B_{t-1} + \frac{T_t}{P_t}, \quad \text{for all } t$$

Yields Euler equation

$$Z_t C_t^{-\sigma} = \beta(1 + i_t) \tilde{E}_t \left[ Z_{t+1} C_{t+1}^{-\sigma} \frac{1}{1 + \pi_{t+1}} \right]$$

and the labor-leisure condition

$$w_t = \Xi C_t^{\sigma}$$

► back

## Final goods producer

There is a representative final good producer that aggregates the intermediate goods  $Y_t(j)$  to a final good  $Y_t$ , according to

$$Y_t = \left( \int_0^1 Y_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (1)$$

with  $\epsilon > 1$ . Nominal profits are given by  $P_t \left( \int_0^1 Y_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}} - \int_0^1 P_t(j) Y_t(j) dj$ , and profit maximization gives rise to the demand for each variety  $j$ :

$$Y_t(j) = \left( \frac{P_t(j)}{P_t} \right)^{-\epsilon} Y_t. \quad (2)$$

Thus, demand for variety  $j$  is a function of its relative price, the price elasticity of demand  $\epsilon$  and aggregate output  $Y_t$ . The aggregate price level is given by

$$P_t = \left( \int_0^1 P_t(j)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}. \quad (3)$$

## Intermediate producers

Intermediate producer of variety  $j$  produces output  $Y_t(j)$  using labor  $H_t(j)$

$$Y_t(j) = H_t(j).$$

When adjusting the price, the firm is subject to a Rotemberg price-adjustment friction.

Per-period profits (in real terms) are given by

$$(1 - \tau_t)P_t(j) \left( \frac{P_t(j)}{P_t} \right)^{-\epsilon} \frac{Y_t}{P_t} - w_t H_t(j) - \frac{\psi}{2} \left( \frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 Y_t + t_t^F(j)$$

Defining  $T_t \equiv 1 - \tau_t$ , it follows that after a linearization of the FOC around the zero-inflation steady state, firm  $j$  sets its price according to

$$\hat{p}_t(j) = \frac{1}{\psi + \epsilon} \left[ \psi \hat{p}_{t-1} + \epsilon \left( \widehat{mc}_t - \hat{T}_t + \hat{p}_t \right) + \beta \psi \tilde{E}_t^j \pi_{t+1}^j \right]$$

► back

## Fiscal policy

The government imposes a sales tax  $\tau_t$  on sales of intermediate goods, issues nominal bonds, and pays lump-sum taxes and transfers  $T_t$  to households and  $t_t^F(j)$  to firms. The real government budget constraint is given by

$$B_t = B_{t-1} \frac{1 + i_{t-1}}{\Pi_t} + \frac{T_t}{P_t} - \tau Y_t + t_t^f.$$

Lump-sum taxes and transfers are set such that they keep real government debt constant at the initial level  $B_{-1}/P_{-1}$ , which I set to zero. [► back](#)



## Subjective expectations of intermediate firms

- ▶ two managers: price setter and forecaster (Adam/Padula (2011))

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 $\Rightarrow \tilde{E}_t^j \pi_{t+1}^j = \tilde{E}_t^j \pi_{t+1} = \tilde{E}_t \pi_{t+1}$ , which leads to equilibrium with  $\pi_t = \pi_t^j$  for all  $j$

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  - ▶ subjective expectations formed as estimated empirically
- ▶ get New Keynesian Phillips Curve with subjective expectations:

$$\pi_t = \beta \tilde{E}_t \pi_{t+1} + \kappa \hat{x}_t + u_t$$

# Households and calibration

## Households' expectations:

- ▶ inflation expectations as estimated empirically
- ▶ consumption solely driven by changes in aggregate wage  $\Rightarrow$  they forecast wages similarly as inflation but no regime changes

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Calibration:

- ▶ calibrate all “non-attention parameters” to standard values in the literature
- ▶ set  $\gamma_{\pi,L} = 0.18$  and  $\bar{\pi} = 3.91\%$  and compute implied information cost  $\tilde{\lambda}$  from

$$\gamma_{\pi,r} = 1 - \frac{\tilde{\lambda}_r}{\sigma_{\pi,r}^2}$$

while increasing  $\sigma_u$  in high-attention regime to match  $\gamma_{\pi,H} = 0.35$  jointly  
 $\Rightarrow \sigma_{u,H} = 1.23\sigma_{u,L}$



# Equilibrium

- ▶ Aggregate supply:

$$\pi_t = \beta \tilde{E}_t \pi_{t+1} + \kappa \hat{x}_t + u_t$$

- ▶ Aggregate demand:

$$\begin{aligned}\hat{x}_t &= \tilde{E}_t \hat{x}_{t+1} - \varphi \left( \tilde{i}_t - \tilde{E}_t \pi_{t+1} - r_t^* \right) \\ \tilde{i}_t &= \rho_i \tilde{i}_{t-1} + (1 - \rho_i) (\phi_\pi \pi_t + \phi_x \hat{x}_t)\end{aligned}$$

+ shocks and expectation formation   ▶ Analytical Example   ▶ back

| Parameter                   | Description                     | Value               |
|-----------------------------|---------------------------------|---------------------|
| $\beta$                     | Discount factor                 | $\frac{1}{1+1/400}$ |
| $\varphi$                   | Interest rate elasticity        | 1                   |
| $\kappa$                    | Slope of NKPC                   | 0.057               |
| $\rho_i$                    | Interest rate smoothing         | 0.7                 |
| $\phi_\pi$                  | Inflation response coefficient  | 2                   |
| $\phi_x$                    | Output gap response coefficient | 0.125               |
| $\rho_u$                    | Shock persistence               | 0.8                 |
| $\sigma_u$                  | Shock volatility                | 0.3%                |
| <u>Attention parameters</u> |                                 |                     |
| $\bar{\pi}$                 | Attention threshold             | 3.91% (annualized)  |
| $\gamma_{\pi,L}$            | Low inflation attention         | 0.18                |
| $\gamma_{\pi,H}$            | High inflation attention        | 0.35                |
| $\gamma_x$                  | Output gap attention            | 0.25                |

## An (hopefully) illustrative example

Consider a stylized version of the model: set  $\tilde{i}_t = \phi_\pi \pi_t$ ,  $\gamma_x = 0$  and  $\tilde{E}_{-1} \hat{x}_0 = 0$

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Focus on first three periods:

0: Steady State

1: Cost-push shock hits:  $u_1 > 0$

2: Shock persists:  $u_2 = u_1 > 0$

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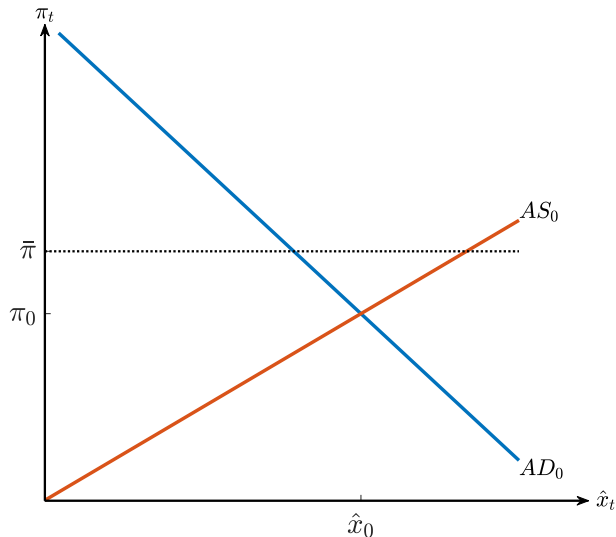
0: Steady State

1: Cost-push shock hits:  $u_1 > 0$

2: Shock persists:  $u_2 = u_1 > 0$

Q: What happens to inflation?

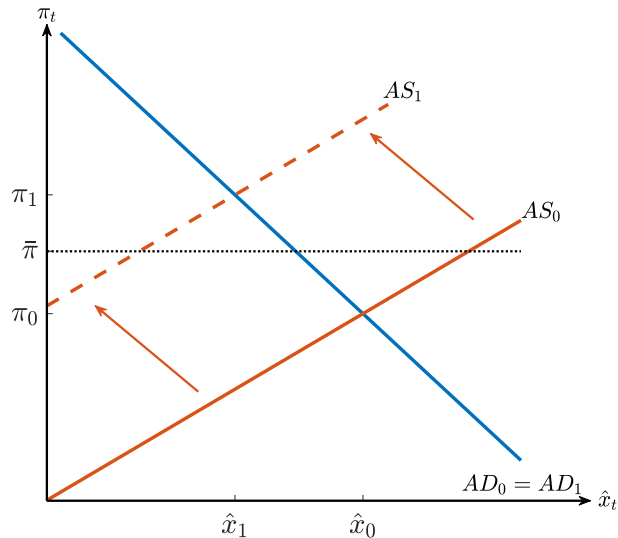
## Period 0: economy in steady state



$$AS_0 : \quad \pi_0 = \frac{\kappa}{1 - \beta\gamma_{\pi,L}} \hat{x}_0$$

$$AD_0 : \quad \pi_0 = -\frac{1}{\phi_{\pi} - \gamma_{\pi,L}} \hat{x}_0$$

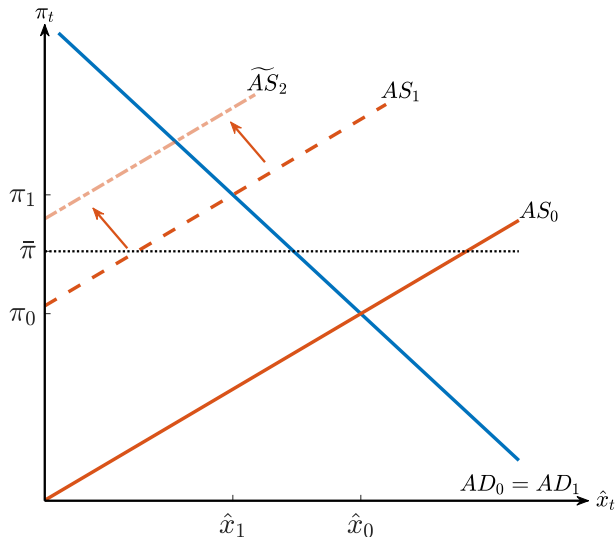
## Period 1: Cost-push shock hits



$$AS_1 : \quad \pi_1 = \frac{\kappa}{1 - \beta\gamma_{\pi,L}} \hat{x}_1 + \frac{1}{1 - \beta\gamma_{\pi,L}} u_1$$

$$AD_1 : \quad \pi_1 = -\frac{1}{\phi_{\pi} - \gamma_{\pi,L}} \hat{x}_1$$

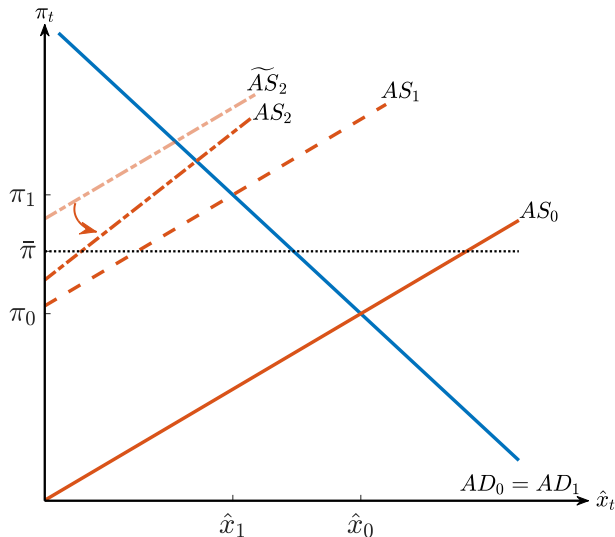
## Period 2: AS further up due to ongoing shock & prior expectations



$$\begin{aligned} \widetilde{AS}_2 : \quad \pi_2 = & \frac{\kappa}{1 - \beta\gamma_{\pi,L}} \hat{x}_2 \\ & + \frac{1}{1 - \beta\gamma_{\pi,H}} u_2 \\ & + \frac{\beta(1 - \gamma_{\pi,H})\gamma_{\pi,L}}{1 - \beta\gamma_{\pi,H}} \pi_1 \end{aligned}$$

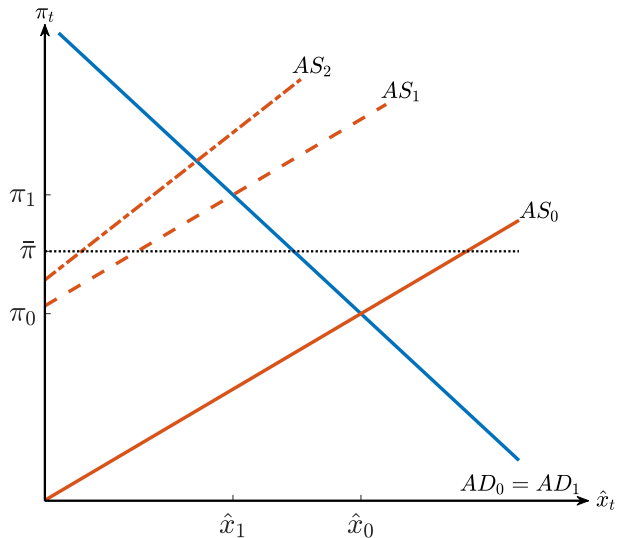


## Period 2: AS becomes steeper due to higher attention

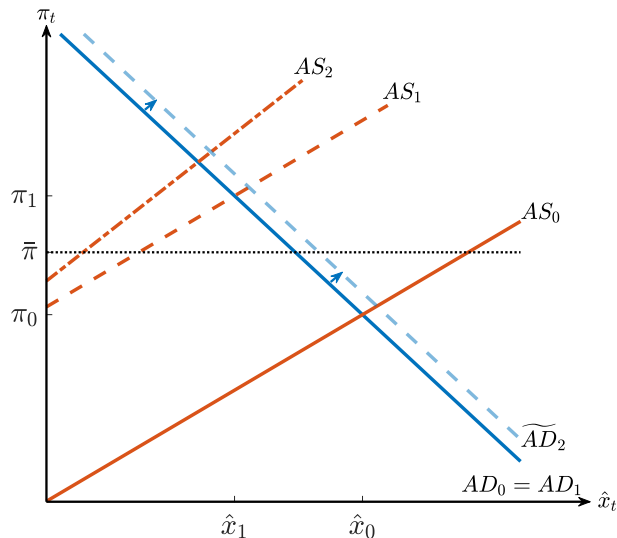


$$\begin{aligned}
 AS_2 : \quad \pi_2 = & \frac{\kappa}{1 - \beta\gamma_{\pi,H}} \hat{x}_2 \\
 & + \frac{1}{1 - \beta\gamma_{\pi,H}} u_2 \\
 & + \frac{\beta(1 - \gamma_{\pi,H})\gamma_{\pi,L}}{1 - \beta\gamma_{\pi,H}} \pi_1
 \end{aligned}$$

## Period 2: What about aggregate demand?

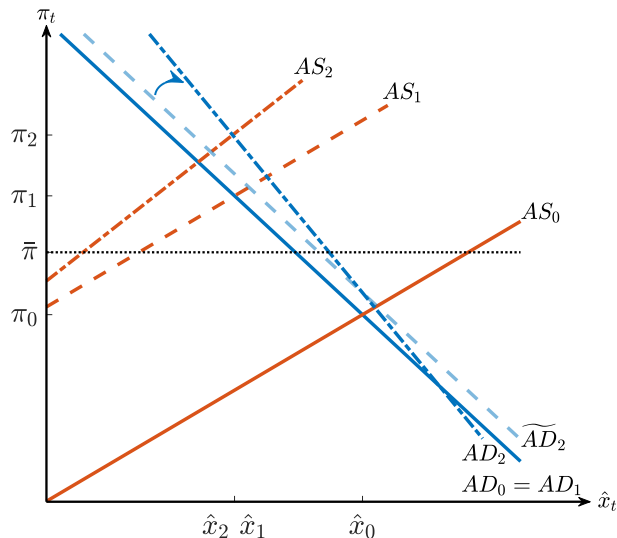


## Period 2: AD shifts out due to positive prior expectations



$$\widetilde{AD}_2 : \quad \pi_2 = - \frac{1}{\phi_\pi - \gamma_{\pi,L}} \hat{x}_2 + \frac{(1 - \gamma_{\pi,H})\gamma_{\pi,L}}{\phi_\pi - \gamma_{\pi,H}} \pi_1$$

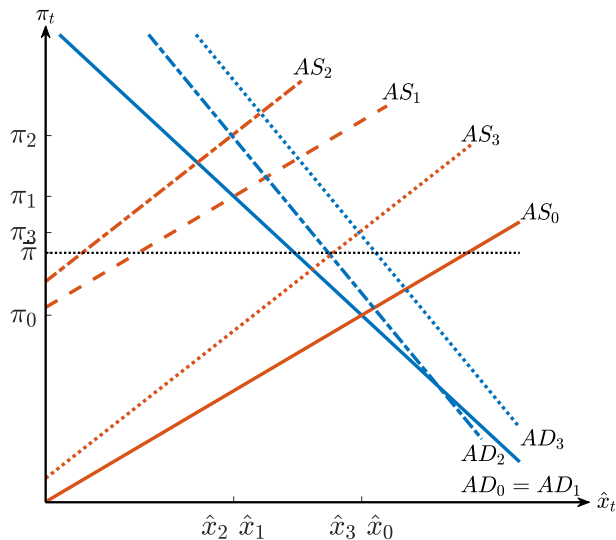
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$$AD_2 : \quad \pi_2 = -\frac{1}{\phi_\pi - \gamma_{\pi,H}} \hat{x}_2 + \frac{(1 - \gamma_{\pi,H})\gamma_{\pi,L}}{\phi_\pi - \gamma_{\pi,H}} \pi_1$$

► Period 3    ► back

## Illustrative example: Period 3

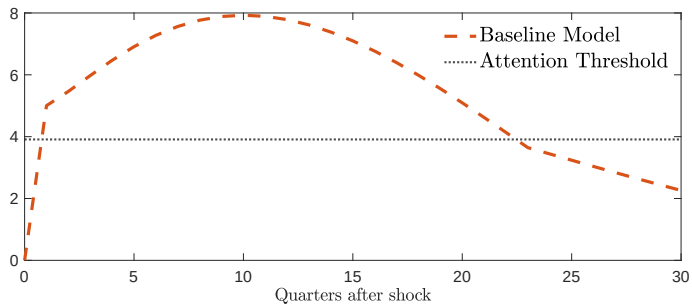


$$AS_3 : \quad \pi_3 = \frac{\kappa}{1 - \beta\gamma_{\pi,H}} \hat{x}_3 + \frac{\beta(1 - \gamma_{\pi,H})}{1 - \beta\gamma_{\pi,H}} \tilde{E}_2 \pi_3$$

$$AD_3 : \quad \pi_3 = - \frac{1}{\phi_{\pi} - \gamma_{\pi,H}} \hat{x}_3 + \frac{1 - \gamma_{\pi,H}}{\phi_{\pi} - \gamma_{\pi,H}} \tilde{E}_2 \pi_3$$

► back

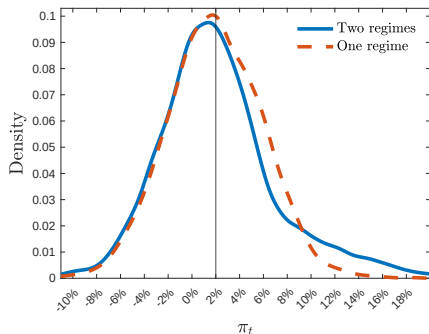
# Demand shocks



► back

# Asymmetry in Inflation Dynamics

- ▶ The attention threshold leads to an **asymmetry** in inflation dynamics  
⇒ heightened risk of high-inflation periods



- ▶ Both models yield similar predictions for median inflation and deflation probabilities
- ▶ average inflation  $> 0$  with 2 regimes  
= 0 with one regime

▶ back

# Implications of different monetary policy rules for central bank losses

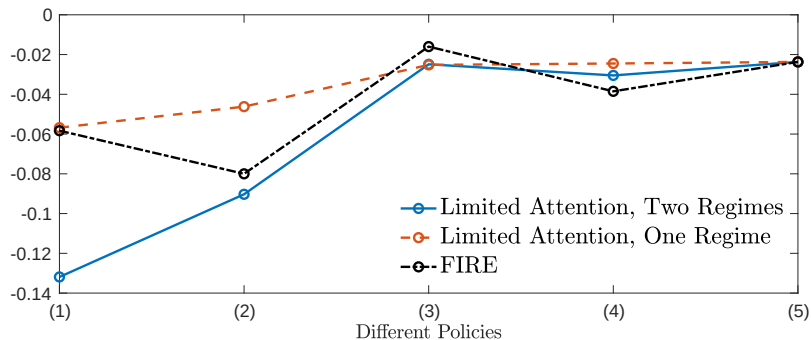
$$\text{Central bank loss} \equiv -\frac{1}{2}E_0 \sum_{t=0}^{\infty} \beta^t [\pi_t^2 + \Lambda \hat{x}_t^2], \text{ with } \Lambda = 0.007$$

Compare welfare implications of different policy rules:

| Nr. | Name                            | Equation                                                                                  |
|-----|---------------------------------|-------------------------------------------------------------------------------------------|
| (1) | Taylor rule with smoothing      | $\tilde{i}_t = \rho_i \tilde{i}_{t-1} + (1 - \rho_i) (\phi_\pi \pi_t + \phi_x \hat{x}_t)$ |
| (2) | Taylor rule without smoothing   | $\tilde{i}_t = \phi_\pi \pi_t$                                                            |
| (3) | Optimal RE commitment policy    | $\pi_t + \frac{\Lambda}{\kappa} (\hat{x}_t - \hat{x}_{t-1}) = 0$                          |
| (4) | Optimal RE discretionary policy | $\pi_t + \frac{\Lambda}{\kappa} \hat{x}_t = 0$                                            |
| (5) | Strict inflation targeting      | $\pi_t = 0$                                                                               |



## Central bank loss

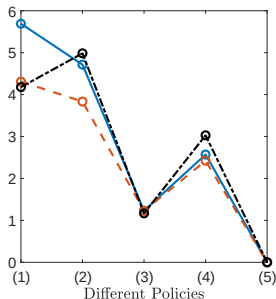


- ▶ Taylor rules lead to larger central bank losses than in other models
  - ▶ especially with interest-rate smoothing

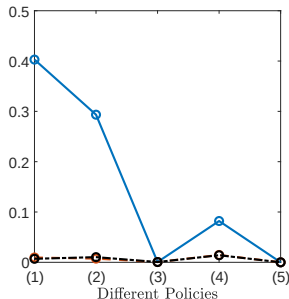
▶ back

# Asymmetry of attention threshold increases average level of inflation

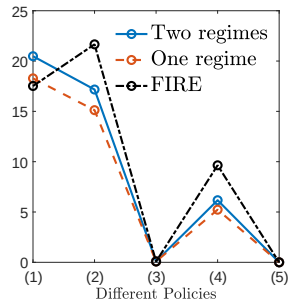
(a) Inflation volatility



(b) Average inflation



(c) Frequency  $H$  regime



► Asymmetry  $\Rightarrow$  average level  $> 0 \Rightarrow$  losses

► back